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Antenna package and image display device including the same

Abstract

An antenna package includes a first antenna device including a first antenna unit, a second antenna device disposed at a level different from that of the first antenna device, the second antenna device including a second antenna unit that has a radiation direction different from that of the first antenna unit, a first circuit board electrically connected to the first antenna unit, a second circuit board electrically connected to the second antenna unit, and a third circuit board electrically and independently connected to the first circuit board and the second circuit board, the third circuit board having at least one antenna driving integrated circuit (IC) chip mounted thereon.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION AND CLAIM OF PRIORITY (1) The present application is a continuation application to International Application No. PCT/KR2021/012452 with an International Filing Date of Sep. 14, 2021, which claims the benefit of Korean Patent Application No. 10-2020-0126493 filed on Sep. 29, 2020 at the Korean Intellectual Property Office, the disclosures of which are incorporated by reference herein in their entirety.

BACKGROUND
1. Field
(1) The present invention relates to an antenna package and an image display device including the same. More particularly, the present invention relates to an antenna package including an antenna device and an intermediate structure and an image display device including the same.
2. Description of the Related Art
(2) As information technologies have been developed, a wireless communication technology such as Wi-Fi, Bluetooth, etc., is combined with an image display device in, e.g., a smartphone form. In this case, an antenna may be combined with the image display device to provide a communication function.
(3) According to developments of a mobile communication technology, an antenna capable of implementing, e.g., high frequency or ultra-high frequency band communication is needed in the display device.
(4) However, when a driving frequency of the antenna increases, a signal interference and a signal loss may also be increased. Further, when a plurality of the antenna devices are included to

improve antenna radiation performance, the signal interference and the signal loss between the plurality of the antenna devices may be further increased.

(5) Thus, a construction of an antenna package capable of preventing the signal interference and the signal loss while implementing high frequency or ultra-high frequency radiation properties using the plurality of the antenna devices.

SUMMARY

(6) According to an aspect of the present invention, there is provided an antenna package having improved radiation property and signaling efficiency.

(7) According to an aspect of the present invention, there is provided an image display device including an antenna package with improved radiation property and signaling efficiency.

(8) (1) An antenna package, including: a first antenna device including a first antenna unit: a second antenna device disposed at a level different from that of the first antenna device, the second antenna device including a second antenna unit that has a radiation direction different from that of the first antenna unit; a first circuit board electrically connected to the first antenna unit: a second circuit board electrically connected to the second antenna unit; and a third circuit board electrically and independently connected to the first circuit board and the second circuit board, the third circuit board having at least one antenna driving integrated circuit (IC) chip mounted thereon.

(9) (2) The antenna package of the above (1), wherein the first antenna unit includes a first radiator radiating in a vertical direction with respect to a top surface of the third circuit board.

(10) (3) The antenna package of the above (2), wherein the second antenna unit includes a second radiator radiating in a horizontal direction with respect to the top surface of the third circuit board.

(11) (4) The antenna package of the above (3), wherein the first radiator has a mesh structure, and the second radiator has a solid structure.

(12) (5) The antenna package of the above (2), wherein the second antenna unit includes a second radiator radiating in a direction perpendicular to the top surface of the third circuit board and opposite to a radiation direction of the first radiator.

(13) (6) The antenna package of the above (1), wherein the antenna driving IC chip includes a first antenna driving IC chip and a second antenna driving IC chip which are separately disposed on the third circuit board, and the first antenna driving IC chip and the second antenna driving IC chip are coupled to the first circuit board and the second circuit board, respectively.

(14) (7) The antenna package of the above (6), further including: a first connector disposed on the first circuit board and electrically connected to the first antenna unit; and a second connector disposed on the second circuit board and electrically connected to the second antenna unit.

(15) (8) The antenna package of the above (7), further including: a third connector disposed on the third circuit board and coupled to the first connector to electrically connect the first antenna unit and the first antenna driving IC chip with each other; and a fourth connector disposed on the third circuit board and coupled to the second connector to electrically connect the second antenna unit and the second antenna driving IC chip with each other.

(16) (9) The antenna package of the above (1), wherein the first circuit board and the second circuit board are flexible printed circuit boards (FPCBs), and the third circuit board is a rigid printed circuit board.

(17) (10) The antenna package of the above (1), wherein the first antenna device further includes a first dielectric layer on which the first antenna unit is disposed, and the second antenna device further includes a second dielectric layer on which the second antenna unit is disposed.

(18) (11) The antenna package of the above (10), wherein the first circuit board is integral with the first dielectric layer, and the second circuit board is integral with the second dielectric layer.

(19) (12) The antenna package of the above (1), further including a circuit device or control device mounted on the third circuit board.

(20) (13) An image display device, including: a display panel; and an antenna package of the above-described embodiments combined with the display panel.

(21) (14) The image display device of the above (13), wherein the first antenna unit includes a first radiator radiating in an upward direction from a top surface of the display panel, and the second antenna unit includes a second radiator radiating in a lateral side direction of the display panel or in a downward direction from a bottom surface of the display panel.
(22) (15) The image display device of the above (14), wherein the third circuit board is disposed under the display panel, and the first circuit board is bent to extend from the top surface of the display panel along a lateral surface and the bottom surface of the display panel to be electrically connected to the third circuit board.
(23) (16) The image display device of the above (14), wherein the second circuit board is bent to extend from the lateral surface of the display panel to the bottom surface of the display panel to be electrically connected to the third circuit board.
(24) According to embodiments of the present invention, a first antenna device and a second antenna device may be included in one antenna package, and the first antenna device and the second antenna device may be electrically connected to a third circuit board on which an antenna driving integrated circuit chip is mounted via a first circuit board and a second circuit board, respectively. For example, a multi-axis directional signal transmission and reception may be implemented in one antenna package, and thus a dual radiation for a high frequency or ultra-high frequency and a broadband signal may be implemented.
(25) In some embodiments, the first antenna device may be disposed on a top surface of a display panel, and the second antenna device may be disposed on a lateral surface or a bottom surface of the display panel. Accordingly, the dual radiation in different directions may be implemented in one antenna package while minimizing signal interference and signal loss. Additionally, a plurality of the antenna devices may be spatially separated, so that a resonance frequency of a band with less signal interference may be selectively used, or a synthesis of a plurality of resonance frequencies may be transmitted and received.
(26) In some embodiments, the first circuit board and the second circuit board may be electrically and independently connected to the third circuit board through a connector. Accordingly, a stable circuit board connection may be implemented without a bonding process or an adhesive process.
(27) The antenna package may be applied to a display device including a mobile communication device capable of transmitting and receiving signals in a 3G, 4G, 5G or higher high-frequency or ultra-high frequency band to improve optical properties such as a transmittance and radiation properties.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIGS. **1** and **2** are schematic plan views illustrating antenna packages in accordance with exemplary embodiments.
(2) FIGS. **3** to **5** are schematic cross-sectional views illustrating an antenna package and an image display device including the same in accordance with exemplary embodiments.
(3) FIG. **6** is a schematic plan view illustrating an image display device in accordance with exemplary embodiments.
DETAILED DESCRIPTION OF THE EMBODIMENTS
(4) According to embodiments of the present invention, an antenna package including a plurality of antenna devices and a circuit board electrically connected to the antenna devices is provided. According to embodiments of the present invention, an image display device including the antenna package is also provided.
(5) The antenna devices may be, e.g., a microstrip patch antenna manufactured in the form of a transparent film, a monopole antenna or a dipole antenna. The antenna devices may be applied,

e.g., to communication devices for high-frequency or ultra-high frequency (e.g., 3G, 4G, 5G or higher) communication. However, the application of the antenna device is not limited to a display device, and the antenna device may be applied to various structures such as a vehicle, a home appliance, an architecture, etc.
(6) Hereinafter, the present invention will be described in detail with reference to the accompanying drawings. However, those skilled in the art will appreciate that such embodiments described with reference to the accompanying drawings are provided to further understand the spirit of the present invention and do not limit subject matters to be protected as disclosed in the detailed description and appended claims.
(7) The terms “upper”, “lower”, “top”, “bottom”, “front”, “rear”, etc., as used herein do not designate absolute positions, but are intended to distinguish different components or relative positions.
(8) FIGS. **1** and **2** are schematic plan views illustrating antenna packages in accordance with exemplary embodiments.
(9) Referring to FIGS. **1** and **2**, the antenna package may include a first antenna device **100**, a second antenna device **200** located at a different level from the first antenna device and having a radiation direction different from that of the first antenna device **100**, a first circuit board **150** electrically connected to the first antenna device **100**, a second circuit board **250** electrically connected to the second antenna device **200**, and a third circuit board electrically and independently connected to the first circuit board **150** and the second circuit board **250**.
(10) The first antenna device **100** and the second antenna device **200** may include a first antenna unit **120** disposed on a first antenna dielectric layer **110** and a second antenna unit **220** disposed on the second antenna dielectric layer **210**, respectively.
(11) The antenna dielectric layers **110** and **210** may include an insulating material having a predetermined dielectric range. For example, the antenna dielectric layers **110** and **210** may include a transparent and flexible resin material capable of being folded. For example, the antenna dielectric layers **110** and **210** may include a polyester-based resin such as polyethylene terephthalate, polyethylene isophthalate, polyethylene naphthalate and polybutylene terephthalate: a cellulose-based resin such as diacetyl cellulose and triacetyl cellulose: a polycarbonate-based resin; an acrylic resin such as polymethyl (meth)acrylate and polyethyl(meth)acrylate: a styrene-based resin such as polystyrene and an acrylonitrile-styrene copolymer: a polyolefin-based resin such as polyethylene, polypropylene, a cycloolefin or polyolefin having a norbornene structure and an ethylene-propylene copolymer: a vinyl chloride-based resin; an amide-based resin such as nylon and an aromatic polyamide: an imide-based resin: a polyethersulfone-based resin: a sulfone-based resin; a polyether ether ketone-based resin; a polyphenylene sulfide resin: a vinyl alcohol-based resin: a vinylidene chloride-based resin; a vinyl butyral-based resin: an allylate-based resin: a polyoxymethylene-based resin: an epoxy-based resin; a urethane or acrylic urethane-based resin; a silicone-based resin, etc. These may be used alone or in a combination of two or more therefrom.
(12) In some embodiments, an adhesive film such as an optically clear adhesive (OCA) or an optically clear resin (OCR) may be included in the antenna dielectric layers **110** and **210**.
(13) In some embodiments, the antenna dielectric layers **110** and **210** may include an inorganic insulating material such as silicon oxide, silicon nitride, silicon oxynitride, glass, etc.
(14) In some embodiments, a dielectric constant of the antenna dielectric layers **110** and **210** may be adjusted in a range from about 1.5 to about 12. When the dielectric constant exceeds about 12, a signal loss through a transmission line may be excessively increased to degrade sensitivity and signaling efficiency in a high or ultra-high frequency band.
(15) The first antenna unit **120** and the second antenna unit **220** may be formed on top surfaces of the first antenna dielectric layer **110** and the second antenna dielectric layer **210**, respectively.
(16) For example, a plurality of the first antenna units **120** may be arranged in an array form along a width direction of the first antenna dielectric layer **110** or the first antenna device **100** to form a

first antenna unit row.
(17) For example, a plurality of the second antenna units **220** may be arranged in an array form along the width direction of the second antenna dielectric layer **210** or the second antenna device **200** to form a second antenna unit row.
(18) The first antenna unit **120** of the first antenna device **100** may include a first radiator **122** and a first transmission line **124**. The second antenna unit **220** of the second antenna device **200** may include a second radiator **222** and a second transmission line **224**.
(19) The radiators **122** and **222** may have, e.g., a polygonal plate shape, and the first transmission line **124** and the second transmission line **224** may extend from one side of the first radiator **122** and the second radiator **222**, respectively. The transmission lines **124** and **224** may be formed as a single member substantially integral with the radiators **122** and **222**.
(20) In exemplary embodiments, the radiators **122** and **222** may provide signal transmission and reception in a high frequency or ultra-high frequency (e.g., 3G, 4G, 5G or higher) band. As a non-limiting example, a resonance frequency of the antenna units **120** and **220** may be in a range from about 24 GHz to 29.5 GHZ, and/or from about 37 GHz to 45 GHz.
(21) In exemplary embodiments, an operable resonance frequency of the radiators **122** and **222** may be controlled by adjusting an area of each radiator.
(22) In some embodiments, the first antenna unit **120** of the first antenna device **100** and the second antenna unit **220** of the second antenna device **200** may have different resonance frequencies. In this case, e.g., transmission and reception of two different types of signals may be implemented in one antenna package, and thus transmission and reception of high-frequency or ultra-high frequency, and broadband signals can be implemented simultaneously.
(23) The first antenna unit **120** and the second antenna unit **220** may further include a first signal pad **126** and a second signal pad **226**, respectively. The first signal pad **126** and the second signal pad **226** may be connected to one end portions of the first transmission line **124** and the second transmission line **224**, respectively.
(24) In some embodiments, the first signal pad **126** and the second signal pad **226** may be provided as integral members with the first transmission line **124** and the second transmission line **224**, respectively, and terminal end portions of the first transmission line **124** and the second transmission line **224** may serve as the first signal pad **126** and the second signal pad **226**, respectively.
(25) In some embodiments, a first ground pad **128** and a second ground pad **228** may be disposed around the first signal pad **126** and the second signal pad **226**, respectively. For example, a pair of the first ground pads **128** may be disposed to face each other with the first signal pad **126** interposed therebetween. A pair of the second ground pads **228** may be disposed to face each other with the second signal pad **226** interposed therebetween.
(26) The first ground pad **128** may be electrically and physically separated from the first transmission line **124** and the first signal pad **126**. The second ground pad **228** may be electrically and physically separated from the second transmission line **224** and the second signal pad **226**. Thus, noises generated when radiation signals are transmitted and received through the signal pads **126** and **226** may be effectively filtered or reduced.
(27) In this case, the first ground pad **128** and the second ground pad **228** may also serve as ground layers for the first radiator **122** and the second radiator **222**, respectively, and vertical radiations may be implemented by the radiators **122** and **222**.
(28) In some embodiments, a separate ground layer may be formed under the radiators **122** and **222**, and a conductive member of the display device to which the antenna device is applied may serve as the ground layer for the radiators **122** and **222**.
(29) The conductive member may include, e.g., a gate electrode of a thin film transistor (TFT) included in a display panel, various wirings such as a scan line or a data line, or various electrodes such as a pixel electrode and a common electrode.

(30) In an embodiment, various structures including a conductive material disposed, e.g., under the display panel may serve as the ground layer. For example, a metal plate (e.g., a stainless steel plate such as a SUS plate), a pressure sensor, a fingerprint sensor, an electromagnetic wave shielding layer, a heat dissipation sheet, a digitizer, etc., may serve as the ground layer.
(31) The antenna unit may include silver (Ag), gold (Au), copper (Cu), aluminum (Al), platinum (Pt), palladium (Pd), chromium (Cr), titanium (Ti), tungsten (W), niobium (Nb), tantalum (Ta), vanadium (V), iron (Fe), manganese (Mn), cobalt (Co), nickel (Ni), zinc (Zn), tin (Sn), molybdenum (Mo), calcium (Ca) or an alloy containing at least one of the metals. These may be used alone or in a combination therefrom.
(32) For example, the antenna units **120 220** may include silver (Ag) or a silver alloy (e.g., silver-palladium-copper (APC)), or copper (Cu) or a copper alloy (e.g., a copper-calcium (CuCa)) to implement a low resistance and a fine line width pattern.
(33) In some embodiments, the antenna units **120** and **220** may include a transparent conductive oxide such as indium tin oxide (ITO), indium zinc oxide (IZO), zinc oxide (ZnOx), indium zinc tin oxide (IZTO), etc.
(34) In some embodiments, the antenna unit **120** and **220** may include a stacked structure of a transparent conductive oxide layer and a metal layer. For example, the antenna unit may include a double-layered structure of a transparent conductive oxide layer-metal layer, or a triple-layered structure of a transparent conductive oxide layer-metal layer-transparent conductive oxide layer. In this case, flexible property may be improved by the metal layer, and a signal transmission speed may also be improved by a low resistance of the metal layer. Corrosive resistance and transparency may be improved by the transparent conductive oxide layer.
(35) The antenna units **120** and **220** may each include a blackened portion, so that a reflectance at surfaces of the antenna units **120** and **220** may be decreased to suppress a visual recognition of the antenna units **120** and **220** due to a light reflectance.
(36) In an embodiment, a surface of the metal layer included in the antenna units **120** and **220** may be converted into a metal oxide or a metal sulfide to form a blackened layer. In an embodiment, a blackened layer such as a black material coating layer or a plating layer may be formed on the antenna units **120** and **220** or the metal layer. The black material or plating layer may include silicon, carbon, copper, molybdenum, tin, chromium, molybdenum, nickel, cobalt, or an oxide, sulfide or alloy containing at least one therefrom.
(37) A composition and a thickness of the blackened layer may be adjusted in consideration of a reflectance reduction effect and an antenna radiation property.
(38) In some embodiments, the first radiator **122** and the first transmission line **124** may include a mesh-pattern structure to improve transmittance. In this case, a dummy mesh electrode (not illustrated) may be formed around the first radiator **122** and the first transmission line **124**.
(39) The first signal pad **126** and the first ground pad **128** may be formed as a solid pattern including the above-described metal or alloy in consideration of a feeding resistance reduction, a noise absorption efficiency and an improvement of a horizontal radiation property.
(40) In some embodiments, the first radiator **122** may have the mesh-pattern structure, and the first transmission line **124**, the first signal pad **126** and the first ground pad **128** may be formed as a solid metal pattern.
(41) In this case, the first radiator **122** may be disposed in a display area of the image display device, and the first transmission line **124**, the first signal pad **126** and the first ground pad **128** may be disposed in a non-display area or a bezel area of the image display device.
(42) In some embodiments, the second radiator **222**, the second transmission line **224**, the second signal pad **226** and the second ground pad **228** disposed on a lateral side or a bottom of the display panel as described below may include a solid structure formed of the above-described metal or alloy in consideration of improving a radiation performance, reducing a feeding resistance, improving a noise absorption efficiency and improving horizontal radiation properties.

(43) In exemplary embodiments, the first circuit board **150** may include a first core layer **160** and first signal wirings **170** formed on a surface of the first core layer **160**. The second circuit board **250** may include a second core layer **260** and second signal wirings **270** formed on a surface of the second core layer **260**. For example, the first circuit board **150** and the second circuit board **250** may be a flexible printed circuit board (FPCB).

(44) In some embodiments, the first antenna dielectric layer **110** may serve as the first circuit board **150**. In this case, the first circuit board **150** (e.g., the first core layer **160** of the first circuit board **150**) may be provided as a member substantially integral with the first antenna dielectric layer **110**. Further, the first signal wiring **170** may be directly connected to the first transmission line **124**, and the first signal pad **126** and the first ground pad **128** may be omitted.

(45) In some embodiments, the second antenna dielectric layer **210** may serve as the second circuit board **250**. In this case, the second circuit board **250** (e.g., the second core layer **260** of the first circuit board **250**) may be provided as a substantially integral member with the second antenna dielectric layer **210**. Further, the second signal wiring **270** may be directly connected to the second transmission line **224**, and the second signal pad **226** and the second ground pad **228** may be omitted.

(46) The first core layer **160** and the second core layer **260** may include a flexible resin such as a polyimide resin, MPI (Modified Polyimide), an epoxy resin, polyester, a cyclo olefin polymer (COP), a liquid crystal polymer (LCP), etc. The first core layer **160** and the second core layer **260** may include internal insulating layers included in the first circuit board **150** and the second circuit board **250**, respectively.

(47) The first and second signal wirings **170** and **270** may serve as, e.g., feeding lines. For example, the first signal wiring **170** and the second signal wiring **270** may be arranged on one surfaces (e.g., surfaces facing the antenna units **120** and **220**) of the first core layer **160** and the second core layer **260**, respectively.

(48) For example, the first circuit board **150** may further include a first cover-lay film formed on the one surface of the first core layer **160** to cover the first signal wirings **170**. For example, the second circuit board **250** may further include a second cover-lay film formed on the one surface of the second core layer **260** to cover the second signal wirings **270**.

(49) The first signal wiring **170** and the second signal wiring **270** may be connected or bonded to the first signal pad **126** of the first antenna unit **120** and the second signal pad **226** of the second antenna unit **220**, respectively. For example, the first cover-lay film and the second cover-lay film of the first circuit board **150** and the second circuit board **250** may be partially removed to expose end portions of the first signal wiring **170** and the second signal wiring **270**. The exposed end portions of the first signal wiring **170** and the second signal wiring **270** may be bonded to the first signal pad **126** and the second signal pad **226**, respectively.

(50) For example, a conductive bonding structure such as an anisotropic conductive film (ACF) may be attached on the first signal pads **126** and the second signal pads **226**. Thereafter, a bonding regions BR of each of the first circuit board **150** and the second circuit board **250** where the end portions of the first antenna signal wirings **170** and the second antenna signal wirings **270** are located may be disposed on the conductive bonding structure.

(51) The bonding regions BR of the first circuit board **150** and the second circuit board **250** may be attached to the first antenna device **100** and the second antenna device **200** through a heat treatment/pressing process, respectively. Accordingly, the first signal wirings **170** and the second signal wirings **270** may be electrically connected to the first signal pads **126** and the second signal pads **226**, respectively.

(52) As illustrated in FIGS. **1** and **2**, the first signal wirings **170** may be independently connected or bonded to each of the first signal pads **126** of the first antenna unit **120**. The second signal wirings **270** may be independently connected or bonded to each of the second signal pads **226** of the second antenna unit **220**. In this case, feeding and control signaling may be implemented from a first

antenna driving integrated circuit (IC) chip **310** and a second antenna driving IC chip **320** to the first antenna unit **120** and the second antenna unit **220**, respectively.

(53) In some embodiments, the predetermined number of the first antenna units **120** may be coupled through the first signal wiring **170**, and the predetermined number of the second antenna units **220** may be coupled through the second signal wiring **270**.

(54) In some embodiments, the first circuit board **150** may be integral with the first dielectric layer **110**, and the second circuit board **250** may be integral with the second dielectric layer **210**. For example, the first core layer **160** may be formed as a substantially integral and unitary member with the first dielectric layer **110**, and the second core layer **260** may be formed as a substantially integral and unitary member with the second dielectric layer **210**. Accordingly, a heating and pressing process such as a separate bonding or adhering process may be omitted, so that signal loss and mechanical damages to the antenna devices **100** and **200** that may occur in the heating and pressing process may be prevented.

(55) In some embodiments, the first circuit board **150** or the first core layer **160** may include a first portion **163** and a second portion **165** having different widths. The second portion **165** may have a width smaller than that of the first portion **163**. The second circuit board **250** or the second core layer **260** may include a third portion **263** and a fourth portion **265** having different widths. The fourth portion **265** may have a width smaller than that of the third portion **263**.

(56) The first portion **163** and the third portion **263** may be provided as main base portions of the first circuit board **150** and the second circuit board **250**, respectively. One end portions of the first portion **163** and the third portion **263** may each include the bonding region BR. For example, the first signal wirings **170** may extend from the bonding region BR toward the second portion **165** on the first portion **163**. For example, the second signal wirings **270** may extend from the bonding region BR toward the fourth portion **265** on the third portion **263**.

(57) The first signal wirings **170** may include a bent portion on the first portion **163**, and the second signal wiring **270** may include a bent portion on the third portion **263** as indicated by dotted circles. Accordingly, the first signal wirings **170** may extend on the relatively narrow second portion **165** with a smaller interval or a higher wiring density than that in the first portion **163**. The second signal wirings **270** may extend on the relatively narrow fourth portion **265** with a smaller interval or a higher wiring density than that in the third portion **263**.

(58) The first circuit board **150** and second circuit board **250** may be electrically connected to a third circuit board **300**.

(59) In some embodiments, the second portion **165** of the first circuit board **100** may serve as a connector coupling portion. For example, the second portion **165** may be bent toward a rear portion of the image display device to be electrically connected to the third circuit board **300**. Accordingly, a circuit connection of the first signal wiring **170** may be easily implemented by using the second portion **165** having a reduced width.

(60) In some embodiments, the fourth portion **265** of the second circuit board **250** may serve as a connector coupling portion. For example, the fourth portion **265** may be bent toward the rear portion of the image display device or may extend at the rear portion to be electrically connected to the third circuit board **300**. Accordingly, a circuit connection of the second signal wirings **270** may be easily implemented by using the fourth portion **265** having a reduced width.

(61) Bonding stability with the first antenna device **100** and the second antenna device **200** may be improved using the first portion **163** and the third portion **263**, respectively, having the increased width. If the antenna units **120** and **220** of the antenna devices **100** and **200** are arranged in an array form, sufficient distribution spaces for the signal wirings **170** and **270** may be achieved by the first portion **163** and the third portion **263**.

(62) In exemplary embodiments, the first circuit board **150** and the third circuit board **300** may be electrically connected to each other through a first-third circuit board connector **180**. The second circuit board **250** and the third circuit board **300** may be electrically connected to each other

through the second-third circuit board connector **280**.
(63) In some embodiments, the first to third circuit board connector **180** and the second to third circuit board connector **280** may be provided as Board to Board (B2B) connectors. The first to third circuit board connector **180** may include a first connector **183** and a third connector **185**, and the second to third circuit board connector **280** may include a second connector **283** and a fourth connector **285**.
(64) For example, the first to third circuit board connector **180** may be mounted through a surface mounting technology (SMT) to be electrically connected to end portions of the first signal wirings **170** on the second portion **165** of the first circuit board **150**. For example, the second-third circuit board connector **280** may be mounted through a surface mounting technology (SMT) to be electrically connected to end portions of the second signal wirings **270** on the fourth portion **265**.
(65) In exemplary embodiments, the third circuit board **300** may be a main board of the image display device or may be a rigid printed circuit board. For example, the third circuit board **300** may include a resin (e.g., epoxy resin) layer impregnated with an inorganic material such as glass fiber as a base insulation layer (e.g., a prepreg), and may include circuit wiring distributed on a surface and within an inside of the base insulation layer.
(66) In exemplary embodiments, at least one antenna driving IC chip may be mounted on the third circuit board **300**.
(67) In some embodiments, two or more antenna driving IC chips may be mounted on the third circuit board **300**. In this case, two or more antenna devices may be included in one antenna package and electrically connected to the antenna driving IC chips. Accordingly, dual radiation may be implemented, and the antenna device may be disposed on a lateral side or a bottom surface of the display panel in addition to a top surface of the display panel, thereby reducing signal interference and signal loss in a high frequency or ultra-high frequency band.
(68) In some embodiments, the antenna driving IC chip may include a first antenna driving IC chip **310** and a second antenna driving IC chip **320** combined with the first circuit board **150** and the second circuit board **250**, respectively. The first antenna driving IC chip **310** and the second antenna driving IC chip **320** may be separately disposed on the third circuit board **300**.
(69) In some embodiments, the third connector **185** may be electrically connected to the first antenna driving IC chip **310** through a first connection wiring **315** included in the third circuit board **300**, and the fourth connector **285** may be electrically connected to the second antenna driving IC chip **320** through a second connection wiring **325** included in the third circuit board **300**.
(70) As indicated by arrows in FIGS. **1** and **2**, the first connector **183** mounted on the first circuit board **150** and the third connector **185** mounted on the third circuit board **300** may be coupled to each other. The second connector **283** mounted on the second circuit board **250** and the fourth connector **285** mounted on the third circuit board **300** may be coupled to each other. For example, the first connector **183** and the second connector **283** may be provided as plug connectors, and the third connector **185** and the fourth connector **285** may be provided as receptacle connectors.
(71) Accordingly, a connection between the first circuit board **150** and the third circuit board **300** may be achieved through the first-third circuit board connector **180**, so that an electrical connection between the first antenna unit **120** and the first antenna driving IC chip **310** may be implemented. Further, a connection between the second circuit board **250** and the third circuit board **300** may be achieved through the second-third circuit board connector **280**, so that an electrical connection between the second antenna driving IC chip **320** and the second antenna unit **220** may be implemented.
(72) Therefore, a feeding/control signals (e.g., a phase signal, a beam tilting signal, etc.) may be applied to the first antenna unit **120** and the second antenna unit **220** from the first antenna driving IC chip **310** and the second antenna driving IC chip **320**, respectively.
(73) An intermediate structure in which the first circuit board **150**, the first to third circuit board connectors **180** and the third circuit board **300** are electrically connected may be formed. An

intermediate structure in which the second circuit board **250**, the second-third circuit board connection connector **280** and the third circuit board **300** are electrically connected may also be formed.
(74) In some embodiments, as described above, the first circuit board **150** and the third circuit board **300** may be electrically coupled to each other, and the second circuit board **250** and the third circuit board **300** may be electrically coupled to each other using the connectors **180** and **280**. Thus, the first and second circuit boards **150** and **250** and the third circuit board **300** may be easily coupled to each other using the connectors **180** and **280** without heating and pressing processes such as an additional bonding process or adhering process.
(75) Therefore, a signal loss in the antenna units **120** and **220** may be prevented while suppressing a dielectric loss due to thermal damages to the substrate caused by the heating and pressing processes, and a resistance increase due to wiring damages.
(76) In some embodiments, the connection between the first circuit board **150** and the third circuit board **300**, and the connection between the second circuit board **250** and the third circuit board **300** may be implemented by the heating and pressing process such as the bonding or adhering process.
(77) In this case, for example, the first antenna driving IC chip **310** and the first signal wirings **170** may be electrically connected through the first connection wirings **315** disposed on the third circuit board **300** to perform the feeding and driving control of the first antenna device **100**. Additionally, for example, the second antenna driving IC chip **320** and the second signal wirings **270** may be electrically connected through the second connection wirings **325** disposed on the third circuit board **300** to perform the feeding and driving control of the second antenna device **200**.
(78) In some embodiments, a circuit device **330** and a control device **340** may be mounted on the third circuit board **300** in addition to the antenna driving IC chips **310** and **320**. The circuit device **330** may include, e.g., a capacitor such as a multilayer ceramic capacitor (MLCC), an inductor, a resistor, etc. The control device **340** may include, e.g., a touch sensor driving IC chip, an application processor (AP) chip, etc.
(79) In exemplary embodiments, the second antenna device **200** may be located at a different level from that of the first antenna device **100**, and may have a radiation direction different from that of the first antenna unit **120**. In this case, dual radiation in different directions may be implemented, and thus signal interference and signal loss between antenna units may be reduced, thereby improving antenna radiation performance while a plurality of antenna devices are included in one antenna package. Additionally, the plurality of antenna devices may be by spatially separated so that a resonance frequency of a band with less signal interference may be selected, or a synthesis of a plurality of resonance frequencies for transmission and reception may be implemented.
(80) In some embodiments, the first radiator **122** of the first antenna unit **120** may radiate in a direction perpendicular to a top surface of the third circuit board **300**.
(81) As illustrated in FIG. **1**, in some embodiments, the second radiator **222** of the second antenna unit **220** may radiate in a horizontal direction with respect to the top surface of the third circuit board **300**. In this case, the dual radiation may be implemented on the top and lateral surfaces of the image display device. For example, the location levels and radiation directions of the antenna devices **100** and **200** are different by an angle of about 90 degrees, so that the signal interference and signal loss can be reduced.
(82) As illustrated in FIG. **2**, in some embodiments, the second radiator **222** of the second antenna unit **220** may radiate in a direction perpendicular to the top surface of the third circuit board **300** and opposite to a radiation direction of the first radiator **122**. In this case, the dual radiation can be implemented on the top and bottom surfaces of the image display device. For example, the location levels and radiation directions of the antenna devices **100** and **200** are different by an angle of about 180 degrees, so that the signal interference and signal loss can be reduced.
(83) In some embodiments, the antenna units **120** and **220** of the antenna devices **100** and **200** may have different resonance frequencies. Accordingly, transmission and reception of a plurality of

resonance frequencies in the high frequency or ultra-high frequency band may be implemented while suppressing the signal interference and signal loss between the antenna devices.
(84) Hereinafter, an image display device including an antenna package according to exemplary embodiments will be described with reference to FIGS. **3** to **5**.
(85) FIGS. **3** to **5** are schematic cross-sectional views illustrating an antenna package and an image display device including the same in accordance with exemplary embodiments.
(86) Referring to FIG. **3**, the first antenna device **100** according to some embodiments may be disposed on a top surface of a display panel **405** of an image display device, and the first radiator **122** of the first antenna unit **120** may radiate over the top surface of the display panel **405**. In FIG. **3**, illustration of the second antenna device **200**, the second circuit board **250**, and the second to third circuit board connectors **280** is omitted for convenience of descriptions.
(87) In some embodiments, the first circuit board **150** electrically connected to the first antenna device **100** may be bent to extend from the top surface of the display panel **405** along lateral and bottom surfaces.
(88) In this case, in some embodiments, the relatively narrow second portion **165** of the first circuit board **150** may be bent so that the first connector **183** may be coupled to the third connector **185** mounted on the third circuit board **300**. Accordingly, the electrical connection with the third circuit board **300** disposed under the display panel **405** may be easily implemented.
(89) Referring to FIG. **4**, the second antenna device **200** according to some embodiments may be disposed on the lateral surface of a display panel **405** of the image display device, and the second radiator **222** of the second antenna unit **220** may radiate in a lateral direction of the display panel **405**. In FIG. **4**, illustration of the first antenna device **100**, the first circuit board **150**, and the first to third circuit board connector **180** is omitted for convenience of descriptions.
(90) In some embodiments, the second circuit board **250** electrically connected to the second antenna device **200** may be bent to extend from the lateral surface of the display panel **405** along the bottom surface.
(91) In this case, in some embodiments, the relatively narrow fourth portion **265** of the second circuit board **250** may be bent so that the second connector **283** may be coupled to the fourth connector **285** mounted on the third circuit board **300**. Accordingly, the electrical connection with the third circuit board **300** disposed under the display panel **405** may be easily implemented.
(92) Referring to FIG. **5**, the second antenna device **200** according to some embodiments may be disposed on the bottom surface of the display panel **405** of the image display device, and the second radiator **222** of the second antenna unit **220** may radiate downward from the bottom surface of the display panel **405**.
(93) In some embodiments, the second circuit board **250** electrically connected to the second antenna device **200** may extend on the bottom surface of the display panel **405**.
(94) In this case, in some embodiments, the second connector **283** mounted on the relatively narrow fourth portion **265** of the second circuit board **250** may be coupled to the fourth connector **285** mounted on the third circuit board **300**. Accordingly, the electrical connection between the second circuit board **250** and the third circuit board **300** may be easily implemented.
(95) FIG. **6** is a schematic plan view illustrating an image display device in accordance with exemplary embodiments. In FIG. **6**, illustration of the second antenna device **200** which may be located on a lateral portion or a rear portion of the image display device **400** is omitted for convenience of descriptions.
(96) Referring to FIG. **6**, an image display device **400** may be implemented in the form of, e.g., a smart phone, and FIG. **6** illustrates a front portion or a window face of the image display device **400**. The front portion of the image display device **400** may include a display area **410** and a peripheral area **420**. The peripheral area **420** may correspond to, e.g., a light-shielding portion or a bezel portion of the image display device.
(97) The first antenna device **100** included in the above-described antenna package may be

disposed toward the front portion of the image display device **400**, and may be disposed on the top surface of the display panel **405**. In some embodiments, the first radiator **122** may be at least partially superimposed over the display area **410**.
(98) In this case, the first radiator **122** may include a mesh-pattern structure, and a reduction of a transmittance due to the first radiator **122** may be prevented. The first signal pad **126** and the first ground pad **128** included in the first antenna unit **120** may be formed as solid metal patterns, and may be disposed in the peripheral area **420** to prevent degradation of an image quality.
(99) The second antenna device **200** included in the above-described antenna package may be disposed toward the lateral or rear portion of the image display device **400**, and may be disposed, e.g., on the side or bottom surface of the display panel **405**.
(100) In this case, the second radiator **222** may have a solid structure, and may provide enhanced radiation performance, noise absorption, and suppressed of signal loss.
(101) In some embodiments, the first circuit board **150** may be bent through, e.g., the second portion **165** to extend toward the third circuit board **300** (e.g., a main board) on which the first antenna driving IC chip **310** is mounted at the rear portion of the image display device **400**.
(102) In some embodiments, the second circuit board **250** may be bent, e.g., from the lateral portion of the image display device **400** through the fourth portion **265** or may extend from the rear portion of the image display device **400** and extend toward the third circuit board **300** (e.g., a main board) on which the second antenna driving IC chip **320** is mounted at the rear portion of the image display device **400**.
(103) In some embodiments, the first circuit board **150** and the third circuit board **300** may be interconnected through the first-third circuit board connector **180**, so that a feeding and an antenna driving control to the first antenna device **100** from the first antenna driving IC chip **310** may be implemented.
(104) In some embodiments, the second circuit board **250** and the third circuit board **300** may be interconnected through the second-third circuit board connector **280**, so that a feeding and an antenna driving control to the second antenna device **200** from the second antenna driving IC chip **320** may be implemented.
(105) As described above, a plurality of antenna devices may be included in one antenna package whole being spatially separated. Accordingly, an antenna package that may reduce the signal interference and signal loss while implementing a multi-axial radiation in the high frequency or ultra-high frequency band may be implemented.

Claims

1. An antenna package comprising: a first antenna device comprising a first antenna unit; a second antenna device disposed at a level different from a level of the first antenna device, the second antenna device comprising a second antenna unit having a radiation direction different from a radiation direction of the first antenna unit; a first circuit board electrically connected to the first antenna unit; a second circuit board electrically connected to the second antenna unit; and a third circuit board electrically and independently connected to the first circuit board and the second circuit board, the third circuit board having at least one antenna driving integrated circuit (IC) chip mounted thereon, wherein the antenna driving IC chip comprises a first antenna driving IC chip and a second antenna driving IC chip which are separately disposed on the third circuit board, and the first antenna driving IC chip and the second antenna driving IC chip are coupled to the first circuit board and the second circuit board, respectively.
2. The antenna package of claim 1, wherein the first antenna unit comprises a first radiator radiating in a vertical direction with respect to a top surface of the third circuit board.
3. The antenna package of claim 2, wherein the second antenna unit comprises a second radiator radiating in a horizontal direction with respect to the top surface of the third circuit board.

4. The antenna package of claim 3, wherein the first radiator has a mesh structure, and the second radiator has a solid structure.
5. The antenna package of claim 2, wherein the second antenna unit comprises a second radiator radiating in a direction perpendicular to the top surface of the third circuit board and opposite to a radiation direction of the first radiator.
6. The antenna package of claim 1, further comprising: a first connector disposed on the first circuit board and electrically connected to the first antenna unit; and a second connector disposed on the second circuit board and electrically connected to the second antenna unit.
7. The antenna package of claim 6, further comprising: a third connector disposed on the third circuit board and coupled to the first connector to electrically connect the first antenna unit and the first antenna driving IC chip with each other; and a fourth connector disposed on the third circuit board and coupled to the second connector to electrically connect the second antenna unit and the second antenna driving IC chip with each other.
8. The antenna package of claim 1, wherein the first circuit board and the second circuit board are flexible printed circuit boards (FPCBs), and the third circuit board is a rigid printed circuit board.
9. The antenna package of claim 1, wherein the first antenna device further comprises a first dielectric layer on which the first antenna unit is disposed, and the second antenna device further comprises a second dielectric layer on which the second antenna unit is disposed.
10. The antenna package of claim 9, wherein the first circuit board is integral with the first dielectric layer, and the second circuit board is integral with the second dielectric layer.
11. The antenna package of claim 1, further comprising a circuit device or control device mounted on the third circuit board.
12. An image display device comprising: a display panel; and an antenna package of claim 1 combined with the display panel.
13. The image display device of claim 12, wherein the first antenna unit comprises a first radiator radiating in an upward direction from a top surface of the display panel; and the second antenna unit comprises a second radiator radiating in a lateral side direction of the display panel or in a downward direction from a bottom surface of the display panel.
14. The image display device of claim 13, wherein the third circuit board is disposed under the display panel; and the first circuit board is bent to extend from the top surface of the display panel along a lateral surface and the bottom surface of the display panel to be electrically connected to the third circuit board.
15. The image display device of claim 13, wherein the second circuit board is bent to extend from the lateral surface of the display panel to the bottom surface of the display panel to be electrically connected to the third circuit board.